The self-attention mechanism can be written as:

$$\begin{aligned}&\text{Self Attention}(X; W_q, W_k, W_v)\\&\quad= \text{softmax}\left(\frac{(XW_q)(XW_k)^\top}{\sqrt{d_k}}\right)(XW_v)\end{aligned}$$

The relational-attention mechanism can be written as:

$$\begin{aligned}&\text{Relational Attention}(X; W_q, W_k, V)\\&\quad= \text{softmax}\left(\frac{(XW_q)(XW_k)^\top}{\sqrt{d_k}}\right)V\end{aligned}$$

The exclusive or (XOR) task can be described by:

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Table 1: Truth table for XOR.

The pairwise transitive inference task can be described by:

Obs.?	x_i	x_j	$x_i > x_j$
Yes	A	B	1
	B	C	1
	B	A	0
	C	B	0
No	A	C	1
	C	A	0

Table 2: Truth table for transitive inference.